Computer science

Course	Software engineering
Learning outcomes	Students will able to: 1. Define various software application domains and remember different process model used in software development. 2. Explain needs for software specifications also they can classify different types of software requirements and their gathering techniques. 3. Convert the requirements model into the design model and demonstrate use of software and user interface design principles. 4. Distinguish among SCM and SQA and can classify different testing strategies and tactics and compare them. 5. Generate project schedule and can construct, design and develop network diagram for different type of Projects. They can also organize different activities of project as per Risk impact factor

Unit	Session	Topics
Unit 1	01	System modeling, The system engineering process, System Procurement
	02	Software process models
		Software specification, Software design and implementation
	03	Software validation Software evolution.
	04	**Requirement Engineering**: Functional and Non-functional requirements, User requirements, System requirement
	05	the software requirements document, Feasibility studies,
	06	Requirements elicitation and analysis
	07	Requirementsvalidation, Requirements management.
Unit 2	08	**Software Design**:

		Architectural design
	09	System structuring, Control models, Modular decomposition
	10	Object Oriented design – Objects and object classes, An object oriented design process;
	11	User interface design– Principles, User interaction, Information presentation, User support
	12	**Verification and Validation**: Planning, Software inspections, Automated static analysis, Clean room software development;
	13	Software testing – Defect testing, Integration testing, Object oriented testing.
Unit 3	14	**Software Management:** Project management – Activities, Planning, Scheduling, Risk management;
	15	Software cost estimation – Productivity, Estimation techniques, Project duration and staffing;
	16	Quality management – Assurance and standards, Planning, Quality control, Software measurements and metrics.